70 Difficult MCQ (A–E) + MCA (I–IV)

Instructions - For **MCQ**: choose **one** best answer (**A–E**). - For **MCA**: each question has Statements **I–IV**. Choose the option (**A–E**) that matches the set of **true** statements. - Some questions include **All of the above** / **None of the above**.

Sets

Q1. Let $U = \{1,2,3,4,5,6,7,8,9,10\}$, $A = \{1,2,3,4,5\}$, $B = \{4,5,6,7\}$. Find $(A \cup B)^c$ (complement relative to U). - A. $\{1,2,3\}$ - B. $\{4,5\}$ - C. $\{8,9,10\}$ - D. $\{6,7,8,9,10\}$ - E. None of the above

Q2. (MCA) Let A, B, C, D be sets. Which statements are **always true**? - I. If $A \subseteq B$, then $A \times C \subseteq B \times C$. - II. If $A \cap B = \varnothing$, then $A \times B = \varnothing$. - III. If A and B are finite, then $|A \times B| = |A||B|$. - IV. $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.

Choose: - A. I, III, IV only - B. I and III only - C. II and III only - D. III and IV only - E. All of the above

Q3. If $|S| = 6$, compute $|\mathcal{P}(S)| - \binom{6}{2}$. - A. 49 - B. 48 - C. 15 - D. 64 - E. None of the above

Q4. Exactly **one** of the following identities is valid for all sets A, B, C. Which one? - A. $A - (B \cup C) = (A - B) \cup (A - C)$ - B. $A - (B \cup C) = (A - B) \cap (A - C)$ - C. $A - (B \cap C) = (A - B) \cap (A - C)$ - D. $A - (B \cap C) = (A - B) - (A - C)$ - E. None of the above

Q5. If $A \subseteq B$, which statement is always true? - A. $A \cup B = A$ - B. $A \cap B = B$ - C. $A - B = \varnothing$ - D. $B - A = \varnothing$ - E. None of the above

Q6. Exactly **one** statement is always true (set complements are relative to a fixed universe U). Which one? - A. $(A \cap B)^c = A^c \cap B^c$ - B. $(A \cup B)^c = A^c \cup B^c$ - C. $(A \cup B)^c = A^c \cap B^c$ - D. $(A^c)^c = \varnothing$ - E. None of the above

Q7. If $|A| = 30$, $|B| = 25$, and $|A \cup B| = 40$, find $|A \cap B|$. - A. 5 - B. 10 - C. 15 - D. 20 - E. None of the above

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Propositional Logic

Q8. The compound proposition $(p \to q) \lor (q \to p)$ is: - A. a tautology - B. a contradiction - C. a contingency - D. logically equivalent to $p \leftrightarrow q$ - E. None of the above

Q9. Which option is logically equivalent to $\neg(p \leftrightarrow q)$? - A. $(p \land q) \lor (\neg p \land \neg q)$ - B. $(p \land \neg q) \lor (\neg p \land q)$ - C. $(p \to q) \land (q \to p)$ - D. $(p \lor q) \land \neg(p \land q)$ - E. Both **B** and **D**

Q10. Simplify $(p \to (q \vee r)) \wedge \neg q$ to an equivalent form **that contains no occurrence of** q except in $\neg q$. - A. $(p \wedge \neg q) \vee r$ - B. $(\neg p \vee q \vee r) \wedge \neg q$ - C. $(\neg p \vee r) \wedge \neg q$ - D. $(p \wedge \neg q) \vee (q \wedge r)$ - E. None of the above

Q11. Argument: Premises $(p \to q), (q \to r), \neg r$. Conclusion $\neg p$. The argument is: - A. valid - B. invalid - C. valid only if q is true - D. valid only if p is false - E. None of the above

Q12. The statement $(p \wedge (p \to q)) \to q$ is: - A. a tautology - B. a contradiction - C. a contingency - D. logically equivalent to $p \to q$ - E. None of the above

Q13. Which statement is logically equivalent to $p \to q$? - A. $q \to p$ - B. $\neg p \to \neg q$ - C. $\neg q \to \neg p$ - D. $p \leftrightarrow q$ - E. None of the above

Q14. Simplify: $\neg\big(p \vee (\neg p \wedge q)\big)$. - A. $\neg p \vee \neg q$ - B. $\neg p \wedge \neg q$ - C. $p \wedge \neg q$ - D. $\neg p \wedge q$ - E. None of the above

Q15. (MCA) Which equivalences are valid for all propositions p, q, r? - I. $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ - II. $p \to q \equiv p \wedge q$ - III. $\neg(p \wedge q) \equiv \neg p \vee \neg q$ - IV. $(p \leftrightarrow q) \equiv (p \to q) \wedge (q \to p)$

Choose: - A. I, III, IV only - B. I and III only - C. II and IV only - D. III only - E. All of the above

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Predicate Logic

Q16. Domain: all people. Let $C(x)$: "x is in this class", $P(x)$: "x passed the test". Translate: "Everyone in this class passed the test." - A. $\forall x\,(C(x) \wedge P(x))$ - B. $\forall x\,(C(x) \to P(x))$ - C. $\exists x\,(C(x) \to P(x))$ - D. $\exists x\,(C(x) \wedge P(x))$ - E. None of the above

Q17. The negation of $\forall x\,(P(x) \to Q(x))$ is: - A. $\forall x\,(P(x) \wedge \neg Q(x))$ - B. $\exists x\,(P(x) \wedge \neg Q(x))$ - C. $\exists x\,(P(x) \to \neg Q(x))$ - D. $\forall x\,(P(x) \to \neg Q(x))$ - E. None of the above

Q18. Domain: integers $\mathbb{Z}$. Exactly **one** statement is true. - A. $\forall x\,\exists y\,(x + y = 0)$ - B. $\exists y\,\forall x\,(x + y = 0)$ - C. $\forall x\,\exists y\,(xy = 1)$ - D. $\exists y\,\forall x\,(xy = 1)$ - E. None of the above

Q19. Let S : $\exists x\,\forall y\,R(x, y)$. Which is equivalent to $\neg S$? - A. $\forall x\,\exists y\,\neg R(x, y)$ - B. $\exists x\,\forall y\,\neg R(x, y)$ - C. $\forall x\,\forall y\,\neg R(x, y)$ - D. $\exists x\,\exists y\,\neg R(x, y)$ - E. None of the above

Q20. Domain: real numbers $\mathbb{R}$. The statement $\forall x\,\exists y\,(y^2 = x)$ is: - A. True - B. False - C. True if the domain is $\mathbb{R}_{\geq 0}$ - D. False only when $x = 0$ - E. None of the above

Q21. "There exists exactly one x such that $P(x)$" is best expressed as: - A. $\exists x\,P(x)$ - B. $\exists x\,(P(x) \wedge \forall y\,(P(y) \to y = x))$ - C. $\forall x\,\exists y\,(P(x) \wedge P(y))$ - D. $\exists x\,\forall y\,(P(x) \leftrightarrow y = x)$ - E. None of the above

Q22. (MCA) Assume the domain is **non-empty**. Which inferences are valid? - I. $\forall x\,P(x) \Rightarrow \exists x\,P(x)$ - II. $\exists x\,P(x) \Rightarrow \forall x\,P(x)$ - III. $\forall x(P(x) \to Q(x))$ and $\forall x\,P(x) \Rightarrow \forall x\,Q(x)$ - IV. $\exists x(P(x) \wedge Q(x)) \Rightarrow (\exists x P(x)) \wedge (\exists x Q(x))$

Choose: - A. I, III, IV only - B. I and III only - C. II and IV only - D. III only - E. All of the above

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Proof

Q23. Best proof method for: “If n is odd, then n^2 is odd.” - A. Proof by contradiction - B. Proof by contrapositive - C. Direct proof - D. Proof by cases - E. None of the above

Q24. Which choice gives **counterexamples** to: $\sqrt{x+y} = \sqrt{x} + \sqrt{y}$ for all real x, y? - A. $(1,1)$ only - B. $(4,5)$ only - C. $(1,1)$ and $(4,5)$ - D. None of the above - E. All of the above

Q25. The contrapositive of “If n^2 is even then n is even” is: - A. If n is even then n^2 is even - B. If n is odd then n^2 is odd - C. If n is odd then n^2 is even - D. If n^2 is odd then n is odd - E. None of the above

Q26. To prove a biconditional statement “A iff B”, it is sufficient to: - A. prove $A \to B$ only - B. prove $B \to A$ only - C. prove both $A \to B$ and $B \to A$ - D. prove $\neg A \to \neg B$ only - E. None of the above

Q27. Which pair of cases forms a complete proof by cases for statements about any integer n? - A. n is even - B. n is odd - C. n is even **or** n is odd - D. n is prime **or** n is composite - E. None of the above

Q28. (MCA) Which proof skeletons correctly establish $P \to Q$? - I. Assume P and derive Q. - II. Assume $\neg Q$ and derive $\neg P$. - III. Assume $P \wedge \neg Q$ and derive a contradiction. - IV. Assume $\neg P$ and derive $\neg Q$.

Choose: - A. I, II, III only - B. I and II only - C. II and IV only - D. III only - E. All of the above

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Functions

Q29. Let $f : A \to B$. Which condition guarantees that $f^{-1} : B \to A$ exists as a **function**? - A. f is injective - B. f is surjective - C. f is bijective - D. f is constant - E. None of the above

Q30. Let $f(x) = 2x - 3$, $g(x) = x^2 + 1$. Compute $(g \circ f)(-1)$. - A. 4 - B. 10 - C. 26 - D. 28 - E. None of the above

Q31. Let $f : \{1,2,3,4\} \to \{a,b,c\}$ be defined by $f(1) = a$, $f(2) = b$, $f(3) = b$, $f(4) = c$. Which is correct? - A. injective and surjective - B. injective only - C. surjective only - D. neither - E. None of the above

Q32. Number of functions from a 5-element set to a 3-element set is: - A. 15 - B. 125 - C. 243 - D. $3^5 - 1$ - E. None of the above

Q33. Number of bijections from a 4-element set to itself is: - A. 16 - B. 24 - C. 256 - D. 12 - E. None of the above

Q34. Let $f(x) = \dfrac{x-1}{x+2}$ (domain $\mathbb{R} \setminus \{-2\}$). Find $f^{-1}(x)$. - A. $\dfrac{x+1}{1-2x}$ - B. $\dfrac{2x+1}{1-x}$ - C. $\dfrac{x+2}{x-1}$ - D. $\dfrac{1-x}{2x+1}$ - E. None of the above

Q35. (MCA) Which statements are true? - I. If f is bijective, then f is injective. - II. If f is bijective, then f is surjective. - III. If f is injective, then $f^{-1} : B \to A$ exists as a function. - IV. If f is surjective, then $f^{-1} : B \to A$ exists as a function.

Choose: - A. I and II only - B. I and III only - C. II and IV only - D. I, II and III only - E. All of the above

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Relations

Q36. Let $A = \{1, 2, 3\}$ and $R = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$. Which properties does R have? - A. reflexive only - B. reflexive and symmetric only - C. reflexive, symmetric, and transitive - D. symmetric and transitive only (not reflexive) - E. None of the above

Q37. For the relation R in Q36, the equivalence class $[1]$ is: - A. $\{1\}$ - B. $\{2\}$ - C. $\{1, 2\}$ - D. $\{1, 2, 3\}$ - E. None of the above

Q38. On $\mathbb{Z}$, define aRb iff $a - b$ is even. How many equivalence classes does R have? - A. 1 - B. 2 - C. infinitely many - D. depends on the subset chosen - E. None of the above

Q39. On $\{1, 2, 3, 4\}$, the relation $\leq$ is best described as: - A. an equivalence relation - B. a strict order - C. a partial order - D. neither - E. None of the above

Q40. Let $A = \{a, b, c\}$ with order (a, b, c). If $R = \{(a,b), (b,c), (c,a), (a,a)\}$, which zero–one matrix represents R? - A. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ - B. $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ - C. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ - D. All of the above - E. None of the above

Q41. (MCA) Let R be a relation on a set X. Which statements are true? - I. R is symmetric **iff** $R = R^{-1}$. - II. If R is transitive, then R^{-1} is transitive. - III. $|R| = |R^{-1}|$. - IV. If R is reflexive, then R^{-1} must be symmetric.

Choose: - A. I, II, III only - B. I and III only - C. II and IV only - D. IV only - E. All of the above

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Matrix

Q42. Let $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix}$. Compute AB. - A. $\begin{bmatrix} 6 & -3 \\ 12 & 3 \end{bmatrix}$ - B. $\begin{bmatrix} 2 & -5 \\ 12 & 3 \end{bmatrix}$ - C. $\begin{bmatrix} 6 & 3 \\ 12 & -3 \end{bmatrix}$ - D. $\begin{bmatrix} -6 & 3 \\ -12 & -3 \end{bmatrix}$ - E. None of the above

Q43. Solve: $2x + y = 5$, $-x + 3y = 4$. - A. $(x, y) = (11/7, 13/7)$ - B. $(x, y) = (13/7, 11/7)$ - C. $(x, y) = (1, 3)$ - D. $(x, y) = (2, 1)$ - E. None of the above

Q44. Let A be 2×3 and B be 3×2. Which statement is correct? - A. Both AB and BA are defined and have the same size. - B. AB is defined, BA is not. - C. BA is defined, AB is not. - D. Both are defined; AB is 2×2 and BA is 3×3. - E. None of the above

Q45. Assume AB is defined. Which statement is always true? - A. $(AB)^T = A^T B^T$ - B. $(AB)^T = B^T A^T$ - C. $(A^T)^T = A^{-1}$ - D. $(A + B)^T = A^T B^T$ - E. None of the above

Q46. Using the matrices in Q42, compute $2A - B$. - A. $\begin{bmatrix} -2 & -5 \\ 7 & -2 \end{bmatrix}$ - B. $\begin{bmatrix} 2 & 5 \\ -7 & 2 \end{bmatrix}$ - C. $\begin{bmatrix} -2 & 5 \\ 7 & 2 \end{bmatrix}$ - D. $\begin{bmatrix} -2 & -5 \\ -7 & -2 \end{bmatrix}$ - E. None of the above

Q47. Does $\begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ have an inverse? - A. Yes, it is invertible - B. No, it is not invertible (determinant is 0) - C. Yes, determinant is 2 - D. Yes, determinant is -2 - E. None of the above

Q48. (MCA) Which statements are true for invertible $n \times n$ matrices? - I. If A is invertible then $\det(A) \neq 0$. - II. If $\det(A) = 0$ then A is not invertible. - III. If A is invertible then $(A^{-1})^{-1} = A$. - IV. If A and B are invertible then $(AB)^{-1} = A^{-1}B^{-1}$.

Choose: - A. I, II, III only - B. I and II only - C. II and IV only - D. III and IV only - E. All of the above

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Counting

Q49. How many distinct permutations of the letters in **MISSISSIPPI** are there? - A. 34650 - B. 36960 - C. 3465 - D. $\dfrac{11!}{10}$ - E. None of the above

Q50. How many ways to choose 5 students from 12 students? - A. 240 - B. 720 - C. 792 - D. 950 - E. None of the above

Q51. How many 6-digit PINs can be formed using digits 0–9 with **no repeated digit**? - A. 10^6 - B. 9^6 - C. $10P4$ - D. $10P6$ - E. None of the above

Q52. How many 4-letter passwords can be formed using uppercase letters (A–Z) if repetition is allowed? - A. $26P4$ - B. 4^{26} - C. 26^4 - D. $4^{26} - 1$ - E. None of the above

Q53. There are 5 IT books and 4 Mathematics books, all with different titles. How many ways to arrange them on a shelf if books of the same topic must be grouped together, **IT on the left, Mathematics on the right**? - A. $9!$ - B. $5! \cdot 4!$ - C. $2 \cdot 5! \cdot 4!$ - D. $5! + 4!$ - E. None of the above

Q54. (MCA) Ten persons have first names from {Alice, Bernard, Charles} and last names from {Lee, McDuff, Ng}. Which statements are true? - I. There are $3 \times 3 = 9$ possible (first name, last name) combinations. - II. With 9 persons, it is guaranteed that at least two share the same full name. - III. With 10 persons, it is guaranteed that at least two share the same full name. - IV. This is an application of the pigeonhole principle.

Choose: - A. I, III, IV only - B. I and III only - C. II and IV only - D. III only - E. All of the above

Q55. (MCA) Which statements are correct? - I. $P(n, r) = \dfrac{n!}{(n-r)!}$ - II. If repetition is allowed and order matters, selecting r objects from n distinct objects gives n^r arrangements. - III. $\binom{n}{r} = \dfrac{n!}{(n-r)!r!}$ - IV. If n objects contain repeats with counts $n_1, \ldots, n_k$ ($\sum n_i = n$), then distinct permutations are $\dfrac{n!}{n_1! \cdots n_k!}$.

Choose: - A. I, II, III only - B. I and II only - C. II and IV only - D. IV only - E. I, II, III and IV

Graph Theory

Q56. A simple undirected graph has 9 vertices and 16 edges. The sum of degrees of all vertices is: - A. 16 - B. 18 - C. 32 - D. 34 - E. None of the above

Q57. A connected undirected graph has vertex degrees $3, 3, 2, 2, 2, 2$. Which is correct? - A. An Euler circuit exists - B. An Euler trail exists but no Euler circuit - C. Neither an Euler trail nor an Euler circuit exists - D. Both an Euler trail and an Euler circuit exist - E. None of the above

Q58. A connected undirected graph has an Euler circuit **iff**: - A. exactly two vertices have odd degree - B. every vertex has even degree - C. the graph is complete - D. the graph is bipartite - E. None of the above

Q59. Consider the path graph $1 - 2 - 3$ with adjacency matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. What is $(A^2)_{1,3}$? - A. 0 - B. 1 - C. 2 - D. 3 - E. None of the above

Q60. In any finite undirected graph, the number of vertices with odd degree is: - A. always odd - B. always even - C. always 0 - D. always equal to the number of edges - E. None of the above

Q61. (MCA) Which statements are true for **undirected** graphs? - I. A graph is bipartite **iff** it contains no odd cycle. - II. Every tree is bipartite. - III. K_3 is bipartite. - IV. $K_{m,n}$ has mn edges.

Choose: - A. I, II, IV only - B. I and II only - C. II and III only - D. III and IV only - E. All of the above

Q62. Weighted undirected graph edges: A–$B = 4$, A–$C = 1$, C–$B = 2$, B–$D = 1$, C–$D = 5$. The shortest-path distance from A to D is: - A. 4 - B. 5 - C. 6 - D. 7 - E. None of the above

Q63. (MCA) Which statements are true for trees (assume $n \geq 2$)? - I. If G is a tree, then G is connected and acyclic. - II. If G is connected and has $n - 1$ edges, then G is a tree. - III. Every tree has at least two leaves. - IV. In a tree, there is a unique simple path between any two vertices.

Choose: - A. I and II only - B. I, II, IV only - C. II, III, IV only - D. I, II, III, IV - E. None of the above

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Trees

Q64. A tree with 15 vertices has how many edges? - A. 13 - B. 14 - C. 15 - D. 16 - E. None of the above

Q65. In a full 3-ary tree (each internal node has exactly 3 children) with 10 internal nodes, the number of leaves is: - A. 20 - B. 21 - C. 22 - D. 31 - E. None of the above

Q66. Maximum number of nodes in a binary tree of height 4 (root at level 0) is: - A. 15 - B. 16 - C. 31 - D. 32 - E. None of the above

Q67. In a rooted tree, the **depth** of a node is: - A. the number of children of the node - B. the number of edges on the unique path from the root to the node - C. the number of nodes in the node's subtree - D. the height of the tree - E. None of the above

Q68. For an unweighted graph, which search guarantees shortest-path distances from a start vertex to all reachable vertices? - A. DFS - B. BFS - C. Both DFS and BFS - D. Neither DFS nor BFS - E. None of the above

Q69. In a full binary tree (each internal node has exactly 2 children), if there are 17 leaves, the number of internal nodes is: - A. 16 - B. 17 - C. 18 - D. 34 - E. None of the above

Q70. (MCA) In any connected simple graph with n vertices: - I. A spanning tree has exactly $n - 1$ edges. - II. Removing any one edge from a spanning tree disconnects it. - III. Adding any one edge to a tree creates exactly one cycle. - IV. Every connected graph has at least one spanning tree.

Choose: - A. I and II only - B. I, II, III only - C. II, III, IV only - D. I, II, III, IV - E. None of the above

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Answer Scheme (Back)

1 C 2 A 3 A 4 B 5 C 6 C 7 C 8 A 9 E 10 C 11 A 12 A 13 C 14 B 15 A 16 B 17 B 18 A 19 A 20 B 21 B 22 A 23 C 24 C 25 B 26 C 27 C 28 A 29 C 30 C 31 C 32 C 33 B 34 B 35 A 36 C 37 C 38 B 39 C 40 A 41 A 42 A 43 A 44 D 45 B 46 A 47 B 48 A 49 A 50 C 51 D 52 C 53 B 54 A 55 E 56 C 57 B 58 B 59 B 60 B 61 A 62 A 63 D 64 B 65 B 66 C 67 B 68 B 69 A 70 D